

Laplace transforms of derivatives

First derivative: $\mathcal{L}\left[\frac{df}{dt}\right] = -f(0) + s\tilde{f}(s)$

Proof:
$$\begin{aligned}\mathcal{L}\left[\frac{df}{dt}\right] &= \int_0^{\infty} e^{-st} \frac{df}{dt} dt \\ &= \left[f(t) e^{-st} \right]_0^{\infty} + s \int_0^{\infty} f(t) e^{-st} dt \\ &= -f(0) + s\tilde{f}(s)\end{aligned}$$

Note: we have assumed that $f(t) e^{-st} \rightarrow 0$ as $t \rightarrow \infty$.

Second derivative: $\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2\tilde{f}(s) - sf(0) - f'(0)$

Proof:
$$\begin{aligned}\mathcal{L}\left[\frac{d^2f}{dt^2}\right] &= \int_0^{\infty} e^{-st} \frac{d^2f}{dt^2} dt \\ &= \left[e^{-st} \frac{df}{dt} \right]_0^{\infty} + s \int_0^{\infty} e^{-st} \frac{df}{dt} dt \\ &= -f'(0) + s \mathcal{L}\left[\frac{df}{dt}\right] \\ &= -f'(0) + s [s\tilde{f}(s) - f(0)]\end{aligned}$$

Solving differential equations by Laplace transforms

- Laplace transforms are often used to solve differential equations when the initial conditions are supplied.

- We use the formulas above to turn a differential equation into an algebraic equation, solve it, and invert the transform.
- Why use Laplace transforms?

① They can make it easier to find a particular integral when the RHS is not zero.

② They go directly to the solution for a given set of initial conditions.

Example : Find the solution of

$$y'' + 4y' + 4y = t^2 e^{-2t}$$

with initial conditions $y(0) = y'(0) = 0$.

- Laplace transform of the LHS:

$$s^2 \tilde{y} - s y(0) - y'(0) + 4s \tilde{y} - 4y(0) + 4 \tilde{y}$$

- Laplace transform of the RHS: $\mathcal{L}[t^2 e^{-2t}]$

- We recall that $\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$

$$\text{so that } \mathcal{L}[t^2] = \frac{2}{s^3}$$

- We now apply the shift theorem

$$\mathcal{L}[e^{-at} f(t)] = \tilde{f}(s+a)$$

to see that $\mathcal{L}[t^2 e^{-2t}] = \frac{2}{(s+2)^3}$

- Since $y(0) = y'(0) = 0$, we have that

$$(s^2 + 4s + 4) \tilde{y} = \frac{2}{(s+2)^3}$$

and $\tilde{y} = \frac{2}{(s+2)^5}$

- To find the solution as a function of t , we take the inverse Laplace transform of

$$\tilde{y}: \mathcal{L}^{-1}[\tilde{y}] = \mathcal{L}^{-1}\left[\frac{2}{(s+2)^5}\right] = 2 \mathcal{L}^{-1}\left[\frac{1}{(s+2)^5}\right]$$

- Again, we use the result $\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$

or $\mathcal{L}^{-1}[1/s^{n+1}] = t^n/n!$

with the shift theorem to find that

$$y = 2 \mathcal{L}^{-1}\left[\frac{1}{(s+2)^5}\right] = \frac{2 t^4 e^{-2t}}{4!} = \frac{t^4 e^{-2t}}{12}$$

Application: damped oscillations

A mass m moves along the x -axis and is attracted towards the origin by a force kx , where $k > 0$ is a constant. A damping force is given by $\gamma dx/dt$, where $\gamma > 0$ is a constant. Describe the motion, given that $x(0) = x_0$ and $x'(0) = v_0$.

The equation of motion is

$$\underbrace{m \frac{d^2 x}{dt^2}}_{\text{mass} \times \text{acceleration}} = \underbrace{-kx - \gamma \frac{dx}{dt}}_{\text{net force}}$$

$$\text{or} \quad \frac{d^2 x}{dt^2} + 2\alpha \frac{dx}{dt} + \omega^2 x = 0$$

$$\text{defining } \alpha = \gamma / 2m, \quad \omega^2 = k/m$$

Using the initial conditions, the Laplace transform of this differential equation is

$$s^2 \tilde{x} - x_0 s - v_0 + 2\alpha (s\tilde{x} - x_0) + \omega^2 \tilde{x} = 0$$

$$\text{and} \quad \tilde{x} = \frac{s x_0 + v_0 + 2\alpha x_0}{s^2 + 2\alpha s + \omega^2}$$

$$= \frac{(s + \alpha) x_0}{(s + \alpha)^2 + \omega^2 - \alpha^2} + \frac{v_0 + \alpha x_0}{(s + \alpha)^2 + \omega^2 - \alpha^2}$$

- We can now identify three cases:

$$\textcircled{1} \quad \omega^2 - \alpha^2 > 0 \quad \Rightarrow \quad \sqrt{\omega^2 - \alpha^2} \in \mathbb{R}$$

Recalling that $\mathcal{L}[\sin(at)] = a / (s^2 + a^2)$,

that $\mathcal{L}[\cos(at)] = s / (s^2 + a^2)$, and using the shift theorem,

$$\mathcal{L}[e^{-\alpha t} f(t)] = \tilde{f}(s + \alpha),$$

we see that

$$\begin{aligned} x = \mathcal{L}^{-1}[\tilde{x}] &= x_0 e^{-\alpha t} \cos(\sqrt{\omega^2 - \alpha^2} t) \\ &+ \frac{(v_0 + \alpha x_0)}{\sqrt{\omega^2 - \alpha^2}} e^{-\alpha t} \sin(\sqrt{\omega^2 - \alpha^2} t) \end{aligned}$$

- The motion is underdamped.

- The frequency of the oscillations is $\sqrt{\omega^2 - \alpha^2} / 2\pi$, and their period is $2\pi / \sqrt{\omega^2 - \alpha^2}$.

- Cases (2) ($\omega^2 - \alpha^2 = 0$) and (3) ($\omega^2 - \alpha^2 < 0$) correspond to critically damped and overdamped motion respectively (see Problems 4).

Partial differential equations (PDEs)

- We will deal with functions of at most four independent variables.
Typically, $u = u(x, y, z, t)$.
- A relationship between u and its partial derivatives $u_x = \partial u / \partial x$, $u_{xy} = \partial^2 u / \partial x \partial y$, ...,

$$f(x, y, z, t; u, u_x, u_{xx}, \dots) = 0$$

where f is some function, is a partial differential equation.

- If f is linear in each of $u, u_x, u_y, \dots$, and the coefficients of these terms are independent of u , the PDE is linear
- Linear PDEs obey the superposition principle: if u_1 and u_2 are solutions, then $u_1 + u_2$ is also a solution.
- The order of the equation is the order of the highest partial derivative.

First-order equations

The method of characteristics

- Focus on the case of two independent variables, $u = u(x, y)$.
- Along a characteristic curve (or characteristic) in the (x, y) plane, the PDE can be integrated directly to give the solution u .
- To use this method, we need the region on which we are solving the PDE to be covered by characteristic curves.

Theory

- We focus on the case

$$A(x, y) \frac{\partial u}{\partial x} + B(x, y) \frac{\partial u}{\partial y} = F(x, y)$$

and consider the family of curves (the characteristics) defined by

$$\frac{dx}{A} = \frac{dy}{B} \quad \text{or} \quad \frac{dy}{dx} = \frac{B}{A}$$

- The element of arc length (squared) along one of these curves is given by

$$ds^2 = dx^2 + dy^2$$

- Since $dx = (A/B) dy$, we can also write

$$ds^2 = \frac{A^2}{B^2} dy^2 + dy^2 = \frac{1}{B^2} (A^2 + B^2) dy^2$$

$$\text{and } ds = \frac{1}{B} \sqrt{A^2 + B^2} dy \quad (1)$$

$$\text{- Similarly, } ds = \frac{1}{A} \sqrt{A^2 + B^2} dx \quad (2)$$

- If we multiply the original PDE by ds , we find

$$A ds \frac{\partial u}{\partial x} + B ds \frac{\partial u}{\partial y} = F ds$$

or, using (1) and (2)

$$A \frac{\sqrt{A^2 + B^2}}{A} dx \frac{\partial u}{\partial x} + B \frac{\sqrt{A^2 + B^2}}{B} dy \frac{\partial u}{\partial y} = F ds$$

$$\sqrt{A^2 + B^2} \left\{ \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right\} = F ds$$

- The term in braces is the total differential of u , du , so

$$\frac{du}{F} = \frac{ds}{\sqrt{A^2 + B^2}}$$

- Using ① and ② again, we also have that

$$\boxed{\frac{du}{F} = \frac{dx}{A} = \frac{dy}{B}}$$

from which the solution u can be found by integration along the characteristics.